Rhovas:

API Design & Enforcement

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Motivation & Philosophy

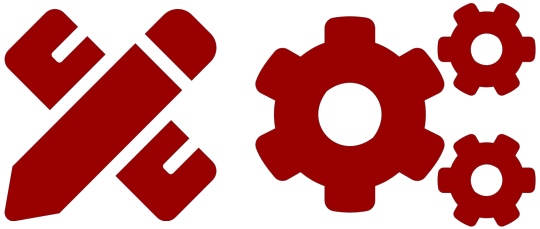
Why make a new language?

Wynaut?

- What's in a Language?
- Storytime (Background)
- Theory of Software APIs
- Three Design Pillars
- Competitive Advantage
- Showcase!

Language

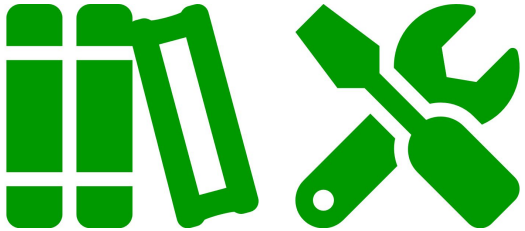
Design



Implementation

Environment

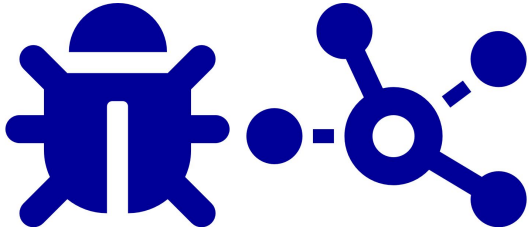
Standard Library



Tooling

Community

Support



Growth

Storytime (Background)

SPONGE
POWERED

Theory of Software APIs

The majority of issues in software development are caused by incorrect assumptions about the API of any given software (including your own).

So... What's the Solution?

Prevent (or at least minimize) incorrect assumptions via:

- **API Design**: Better architecture, consistency with conventions, etc.
- **Enforcement**: Compile-time/runtime - maximize feedback loop.
- **Documentation**: Contracts that can't be enforced at compile-time.

Overall, documentation is the most important:

- All software has a degree of inherent complexity.
- Undocumented behavior is **indistinguishable** from a potential bug.

Example: % Operator

C++/Java (`rem`)

```
7 % 2 == 1
-7 % 2 == -1
```

Ruby/Python (`mod`)

```
7 % 2 == 1
-7 % 2 == 1
```

Three Design Pillars

(of Rhovas)

Correctness Maintainability Performance

Maintainability › Correctness › Performance

Correctness > Performance:

- Performant but not correct: Fails to achieve desired functionality.
- Correct but not performant: Can work to improve performance.

Maintainability > Correctness:

- Correct but not maintainable: Not resilient to specification changes.
- Maintainable but not correct: Can be easily adapted to be correct.
- Also makes it easier to improve performance.

Still requires evaluating tradeoffs for each scenario (e.g. caching).

Competitive Advantage

Key features in Rhovas centered around API Design:

- **Contracts**: Defining preconditions, postconditions, and invariants.
- **Mutability**: Restrict the mutability of shared objects.
- **Error Handling**: Unify expected/unexpected error conditions.
- **API Compatibility**: Identify API breakages and perform migrations.
- **Embedded DSLs**: Use the best language for the problem at hand.

Note: These are all in various stages of being implemented.

- Help wanted ;)

Showcase!

You can talk the talk, but can you walk the walk?

https://rhovas.dev

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